

Case Study: Edge-Computing for Real-Time Traffic Monitoring

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A. Introduction and Objectives

The Liverpool Smart Pedestrians project was made to help city planners get a better grip on how people actually use urban spaces. The goal was to build a sensor system that was cheap and scalable, but most importantly, didn't creep people out or violate their privacy. Before this, planners mostly used manual counts or high-bandwidth CCTV, which are either too slow or too expensive. This project aimed to solve those "data silo" and bandwidth issues by processing everything locally at the source.

B. Methodology

The team used a design science approach, basically building and testing the sensors in a loop to see what worked best. They had some real world constraints to deal with: the devices had to be small enough to mount on lamp posts and tough enough to survive the weather. A huge part of their methodology was "privacy by design." They made sure the sensor processed the video

frames at the edge and deleted them immediately—only sending the anonymous count data (like "10 people moved North") to the cloud.

C. Technology and Implementation

For the hardware, they used the NVIDIA Jetson TX2. I can see why it's a solid piece of gear that puts a lot of GPU power into a small footprint. For the software, they went with YOLO V3. Because YOLO (You Only Look Once) sees the whole frame at once, it's fast enough to handle real-time traffic. This "edge computing" setup is a big deal because it stops the lag you'd get if you tried to send all that heavy video data to a remote server to be processed.

D. Validation and Performance

The researchers tested the system to see how it handled accuracy and speed. Even when the lighting was bad, YOLO V3 was surprisingly accurate at spotting pedestrians. They used the SORT algorithm to track individuals so they didn't get counted twice as they moved. Even though the Jetson TX2 was running at high utilization, the system stayed stable, which proves you don't need a giant server room to run decent AI analytics.

E. Real-World Applications

The study looked at an indoor evacuation and an outdoor street deployment in Liverpool. The indoor test was cool because it showed exactly where bottlenecks happen during emergencies, which is vital for safety design. The outdoor test showed the sensors could actually survive the elements. Connecting this to our class modules, these sensors act as Edge Gateways, filtering out all the "noise" and only sending the important data back to the city's main dashboard.

F. Challenges and Future Work

The biggest issues were "occlusion," basically, when a person is hidden behind a pole or another person, and dealing with heavy rain. Since this was written in 2019, the tech has moved on a lot. If I were building this today for a 2026 project, I'd use the NVIDIA Jetson Orin and YOLOv11. These are much faster and more power-efficient, which would solve a lot of the hardware strain the original team faced during their testing.

G. Personal Evaluation

I think the project was a win, especially for showing that we can have smart cities without "Big Brother" watching everything. To make it better, I'd suggest looking into solar power so the units are totally self sufficient and don't need to be plugged into the grid. Also, using Federated Learning would be a great way to let the sensors get smarter over time without ever having to share the actual video data. It's a great start for making cities more responsive to the people living in them.

References

Barthélemy, Johan, et al. "Edge-Computing Video Analytics for Real-Time Traffic Monitoring in a Smart City." *MDPI*, Multidisciplinary Digital Publishing Institute, 2 May 2019, doi.org/10.3390/s19092048.